

Sequential Monitoring for Structural Breaks in Heterogeneous Panels: Design, Derivations, and Computational Details

Abstract

This note describes a Monte Carlo study of a real-time structural break monitoring procedure for heterogeneous panels. The procedure works in three steps: (1) each series is pre-filtered by an autoregression estimated on a calibration sample; (2) the filtered residuals are aggregated across series via $N(N - 1)$ separate (unpooled) pairwise regressions, one OLS fit per ordered pair of series; (3) the cumulative sum of squares (CUSQ) of the aggregate residual is compared period by period against a boundary derived from the Brownian bridge, following Wang and Hsiao (2013). The note gives the exact data generating processes, a closed-form derivation showing that the $N(N - 1)$ -pair aggregation, even though every pair has its own fitted slope, can be evaluated exactly as a weighted cross-sectional average without ever materializing the $N(N - 1)$ residual series, the two-scenario Monte Carlo design (size under no break, power and detection delay under a known break), and two computational improvements — FFT-based fractional differencing and vectorized AR residuals — that make the full 1.9×10^6 -replication design feasible without changing any statistic. In both scenarios, every replication's single stopping time is read off twice, at no extra simulation cost: once as a *sequential* statistic (whether the monitor ever alarms over the complete horizon, judged against the true break date in the power scenario) and once as a *pointwise* statistic (whether the monitor has alarmed by six fixed, pre-specified calendar dates); the note shows that these answer different questions and must be interpreted, and benchmarked, separately. A companion design adds four targeted power experiments — common-factor and subgroup-specific volatility breaks, and common-factor and subgroup-specific loading breaks — built on the same AR-filtering, pairwise-aggregation, and CUSQ machinery.

1 Setup

We observe a panel of N time series $\{Y_{i,t}\}$ for $i = 1, \dots, N$ and $t = 1, \dots, T$. The first M periods form the calibration sample, assumed to be free of breaks. From period $M + 1$ onward the monitor checks, one period at a time, whether the panel has shifted.

The null and alternative hypotheses are

$$H_0 : \text{no break for any } t > M,$$

H_1 : all N series shift by $\delta \neq 0$ at some time $\tau > M$.

Sections 2–5 describe the four steps of the procedure: data generation, AR filtering, cross-sectional aggregation, and the CUSQ test. Section 7 describes the Monte Carlo design, Section 8 describes four additional targeted power experiments built on the same monitoring procedure, and Section 9 documents the computational improvements.

2 Data Generating Processes

2.1 Panel series

Each series follows an ARFIMA(p_i, d_i, q_i) model,

$$\Phi_i(L)(1-L)^{d_i}Y_{i,t} = \Theta_i(L)\varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim \text{i.i.d. } N(0, 1),$$

where $\Phi_i(L) = 1 - \phi_i L$ and $\Theta_i(L) = 1 + \theta_i L$ are at most first-order polynomials and $(1-L)^{d_i}$ is the fractional differencing operator. Two panel sizes are used.

$N = 10$. Five pairs of series, each pair sharing the same parameters (Table 1).

Table 1: DGP for $N = 10$.				
Series	Model	ϕ_i	d_i	θ_i
1, 2	ARMA(1, 1)	0.8	0	0.5
3, 4	ARMA(1, 1)	0.5	0	0.3
5, 6	ARFIMA(0, d , 0)	0	0.4	0
7, 8	ARFIMA(1, d , 1)	0.1	0.3	0.8
9, 10	ARFIMA(1, d , 0)	0.7	0.4	0

$N = 30$. A random-coefficient panel divided into four blocks (Table 2). The coefficients are drawn independently for each series and for each Monte Carlo replication. This means the simulation averages over both the randomness in the coefficients and the randomness in the innovations, giving a measure of average performance across the whole class of DGPs described in the table rather than performance at one fixed parameter vector.

2.2 Common factors

Four common factors are generated independently of the panel series (Table 3). They cover i.i.d. noise, a near-unit-root AR(1), and two long-memory processes.

Table 2: DGP for $N = 30$ (random coefficients, redrawn each replication).

Series	Model	ϕ_i	d_i	θ_i
1–15	ARMA(1, 1)	$U(0, 0.5)$	0	0.6
16–20	ARMA(1, 1)	$U(0.7, 0.9)$	0	0.6
21–25	ARFIMA(0, d , 0)	0	$U(0, 0.5)$	0
26–30	ARFIMA(1, d , 1)	$U(0, 0.4)$	$U(0, 0.5)$	$U(0, 0.3)$

Table 3: Common factor DGPs.

Factor	Model	ϕ	d
1	i.i.d. $N(0, 1)$	0	0
2	AR(1)	0.99	0
3	ARFIMA(0, d , 0)	0	0.20
4	ARFIMA(0, d , 0)	0	0.45

2.3 Observed data and break injection

The observed series add a common factor with loading $\lambda = 0.5$:

$$Y_{i,t}^* = Y_{i,t} + 0.5 f_t, \quad i = 1, \dots, N. \quad (1)$$

Under H_1 , a level shift of size δ is added at time τ :

$$Y_{i,t}^{**} = Y_{i,t}^* + \delta \cdot \mathbf{1}\{t \geq \tau\}, \quad i = 1, \dots, N. \quad (2)$$

The break time τ is specified in Section 7.2.

3 AR Pre-Filtering

For each series i , we fit an AR(p) model on the calibration sample $t = 1, \dots, M$, where

$$p = \max\left(3, 3 \left\lfloor \frac{M-1}{100} \right\rfloor\right). \quad (3)$$

The OLS estimate from the calibration sample is

$$\hat{\beta}_{AR} = \arg \min_{b \in \mathbb{R}^p} \sum_{t=p+1}^M \left(Y_{i,t} - b'(Y_{i,t-1}, \dots, Y_{i,t-p})' \right)^2. \quad (4)$$

This estimate is fixed after period M and never updated. For all $t = p+1, \dots, T$, the one-step-ahead residual is

$$\hat{e}_{i,t} = Y_{i,t} - \hat{\beta}_{AR}'(Y_{i,t-1}, \dots, Y_{i,t-p})'. \quad (5)$$

The first seven observations are discarded as a burn-in. Fixing the AR coefficients after calibration is the key sequential monitoring requirement: parameters are estimated from clean data and then held constant, so the monitoring period residuals are not used to update the model.

4 Cross-Sectional Aggregation

4.1 Unpooled pairwise regression

Let $\hat{e}_{i,t}$ denote the AR residuals for $i = 1, \dots, N$ and $t = 1, \dots, T$. For every ordered pair (i, j) with $i \neq j$ (there are $N(N-1)$ such pairs), we fit a *separate* calibration-period regression

$$\hat{e}_{i,t} = \alpha_{ij} + \beta_{ij} \hat{e}_{j,t} + u_{i,j,t}, \quad t = 1, \dots, M. \quad (6)$$

Unlike a pooled specification, the $N(N-1)$ regressions are *not* stacked into a single sample: each pair (i, j) has its own OLS estimate $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$, computed using only that pair's M calibration observations. As with the AR coefficients of Section 3, $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$ are estimated once from the calibration sample and then held fixed for every $t = 1, \dots, T$. For all t , the residual from pair (i, j) 's regression is

$$U_{i,j,t} = \hat{e}_{i,t} - \hat{\alpha}_{ij} - \hat{\beta}_{ij} \hat{e}_{j,t}. \quad (7)$$

The aggregate signal used in the CUSQ test is, as before, the average of $U_{i,j,t}$ over all ordered pairs:

$$\bar{e}_t \equiv \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^N U_{i,j,t}, \quad t = 1, \dots, T. \quad (8)$$

As stated, computing (8) requires fitting all $N(N-1)$ pairs and forming all $N(N-1)$ residual sequences — an $O(N^2M)$ estimation step followed by an $O(N^2T)$ evaluation step. The next two propositions show that every $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$, and hence $\bar{e}_t$, can be obtained in closed form at a total cost of $O(N^2M)$ (estimation) plus $O(NT)$ (evaluation), without ever materializing the $N(N-1)$ residual series and without any approximation.

4.2 Closed form for $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$

Define the per-series calibration mean

$$\mu_i \equiv \frac{1}{M} \sum_{t=1}^M \hat{e}_{i,t}, \quad i = 1, \dots, N, \quad (9)$$

and the demeaned residual, held fixed and applied to *every* t (not only the calibration sample),

$$f_{i,t} \equiv \hat{e}_{i,t} - \mu_i, \quad t = 1, \dots, T. \quad (10)$$

Let $d_{i,t} \equiv f_{i,t}$ for $t \leq M$ denote the calibration-period values, and define the $N \times N$ Gram matrix of calibration-period centered residuals

$$S_{ij} \equiv \sum_{t=1}^M d_{i,t} d_{j,t}, \quad i, j = 1, \dots, N. \quad (11)$$

Proposition 1. *The single-pair OLS estimates from (6) are, for $i \neq j$,*

$$\hat{\beta}_{ij} = \frac{S_{ij}}{S_{jj}}, \quad \hat{\alpha}_{ij} = \mu_i - \hat{\beta}_{ij} \mu_j. \quad (12)$$

Proof. For a fixed pair (i, j) , (6) is an ordinary simple linear regression of $\hat{e}_{i,t}$ on $\hat{e}_{j,t}$ over $t = 1, \dots, M$, with no pooling across pairs. Its OLS slope is the ratio of the sample covariance of $(\hat{e}_{i,t}, \hat{e}_{j,t})$ to the sample variance of $\hat{e}_{j,t}$: $\sum_t d_{i,t} d_{j,t} / \sum_t d_{j,t}^2 = S_{ij} / S_{jj}$. The intercept follows from the standard simple-regression identity $\hat{\alpha}_{ij} = \mu_i - \hat{\beta}_{ij} \mu_j$. $\square$

4.3 Closed form for $\bar{e}_t$

Define the cross-sectional sum of the centered residuals, $F_t \equiv \sum_{i=1}^N f_{i,t}$, and, for each j , the column sum of the fixed pairwise slopes excluding the diagonal,

$$c_j \equiv \sum_{i \neq j} \hat{\beta}_{ij}, \quad j = 1, \dots, N. \quad (13)$$

Proposition 2. *For every $t = 1, \dots, T$,*

$$\bar{e}_t = \frac{(N-1)F_t - \sum_{j=1}^N c_j f_{j,t}}{N(N-1)}. \quad (14)$$

Proof. By (12), $\hat{\alpha}_{ij} + \hat{\beta}_{ij} \mu_j = \mu_i$, so $U_{i,j,t} = \hat{e}_{i,t} - \hat{\alpha}_{ij} - \hat{\beta}_{ij} \hat{e}_{j,t} = f_{i,t} - \hat{\beta}_{ij} f_{j,t}$. Expanding (8):

$$\bar{e}_t = \frac{1}{N(N-1)} \sum_{i \neq j} (f_{i,t} - \hat{\beta}_{ij} f_{j,t}) = \frac{1}{N(N-1)} \left[\sum_{i \neq j} f_{i,t} - \sum_{i \neq j} \hat{\beta}_{ij} f_{j,t} \right].$$

In the first sum, $f_{i,t}$ is added once for every $j \neq i$, i.e. $N-1$ times for each i , so $\sum_{i \neq j} f_{i,t} = (N-1)F_t$. In the second sum, regrouping by j gives $\sum_{i \neq j} \hat{\beta}_{ij} f_{j,t} = \sum_j f_{j,t} \sum_{i \neq j} \hat{\beta}_{ij} = \sum_j c_j f_{j,t}$. Combining the two terms gives (14). $\square$

Remark 1. *Unlike under pooling, the $N(N-1)$ pairwise regressions here do not share a common slope, so $\bar{e}_t$ is no longer a simple affine map of the plain cross-sectional mean: each series j contributes through $f_{j,t}$ with its own pair-specific weight c_j . (14) is nonetheless exact and never requires forming the $N(N-1)$ residual series. The cost is $O(N^2M)$ to form S (and hence every $\hat{\beta}_{ij}$ and c_j) once from the calibration sample, plus $O(NT)$ to evaluate $\bar{e}_t$ for the full sample — versus $O(N^2T)$ for a direct, pair-by-pair evaluation. For $N = 30$ this still replaces an $N(N-1) = 870$ -pair construction with one 30×30 Gram matrix and a single weighted cross-sectional average.*

5 The CUSQ Statistic and Monitoring Boundary

5.1 Statistic

Let $C_l = \sum_{t=1}^l \bar{e}_t^2$ be the cumulative sum of squares of $\bar{e}_t$. Following Wang and Hsiao (2013) (p. 5), the centered CUSQ for $n \geq m$ is

$$D_n \equiv \frac{C_n}{C_m} - \frac{n-2}{m-2}, \quad (15)$$

and the detector is

$$T_n \equiv \sqrt{\frac{m}{2}} D_n = \sqrt{\frac{m}{2}} \left(\frac{C_n}{C_m} - \frac{n-2}{m-2} \right). \quad (16)$$

This is equivalent to $T_n = m^{-1/2} \hat{S}_n$ in the notation of Wang and Hsiao (2013), where $\hat{S}_n = 2^{-1/2} m \hat{D}_n$.

5.2 Limiting distribution

Under H_0 , Wang and Hsiao (2013) (Theorem 1) show that

$$T_n = m^{-1/2} \hat{S}_{[m\tau]} \Rightarrow W^0(\tau), \quad \tau = \frac{n}{m} \geq 1, \quad (17)$$

where $W^0(\tau) = W(\tau) - \tau W(1)$ is a Brownian bridge and $W(\tau)$ is a standard Wiener process.

5.3 Monitoring boundary

Following Chu et al. (1996), the crossing probability of $W^0(\tau)$ satisfies (Wang and Hsiao, 2013, eq. (5)):

$$P\{|W^0(\tau)| \geq (\tau-1)^{-1}[\tau(a^2 + \ln \tau)]^{1/2} \text{ for some } \tau \geq 1\} = 2[1 - \Phi(a)] + a\phi(a), \quad (18)$$

where Φ and ϕ are the standard-normal CDF and PDF. Setting the right-hand side equal to α and solving numerically gives

$$a_{10\%}^2 = 6.2514, \quad a_{5\%}^2 = 7.8147, \quad a_{1\%}^2 = 11.3449. \quad (19)$$

Wang and Hsiao (2013) state $a_{5\%}^2 = 7.78$ (rounded to two decimal places). Discretising λ as n/m yields the boundary function (Wang and Hsiao, 2013, p. 6):

$$g\left(\frac{n}{m}; a_\alpha^2\right) = \frac{n-m}{m} \left[\frac{n}{n-m} \left(a_\alpha^2 + \ln \frac{n}{n-m} \right) \right]^{1/2}. \quad (20)$$

6 Sequential Decision Rule

At each period $n > m$, compute T_n from (16) and compare it to the boundary $g(n/m; a_\alpha^2)$ from (20). The procedure declares a break at the first period where the statistic crosses the boundary:

$$\hat{\tau}_\alpha \equiv \inf\{n > m : |T_n| \geq g(n/m; a_\alpha^2)\}, \quad (21)$$

with $\hat{\tau}_\alpha = +\infty$ if no crossing occurs before T . The AR coefficients $\hat{\beta}_{AR}$, the pairwise regression coefficients $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$ for every $i \neq j$, and the per-series means μ_i are all fixed from the calibration period and never updated after $t = m$.

7 Monte Carlo Design

The total sample length used in each replication is

$$T_{\text{total}} = 10M + 10, \quad (22)$$

which provides a monitoring window of $9M$ periods beyond the calibration sample. Two scenarios are run.

7.1 Scenario A: size ($\delta = 0$)

No break is injected. Each replication r produces exactly one random object, the stopping time $\hat{\tau}_\alpha^{(r)} \in (M, \infty]$ from (21). Two distinct quantities are computed from it, at no extra simulation cost, and they must not be conflated: a *sequential* statistic, which asks only whether the monitor ever alarms over the complete horizon, and a *pointwise* statistic, which asks whether the monitor has alarmed by a specific, fixed evaluation date. The two answer different questions and, in general, take different numerical values for the same set of replications.

Remark 2 (Sequential vs. pointwise). *A sequential statistic is evaluated on the stopping time $\hat{\tau}_\alpha^{(r)}$ itself, with no reference to any particular calendar date: it is a single yes/no outcome per replication, “did the monitor ever cross the boundary, anywhere in $(M, T_{\text{total}}]$,” and it is the quantity that corresponds to the test’s classical Type-I error (size) or, under a break, to its overall probability of sounding an alarm at all. A pointwise statistic instead fixes a horizon length k in advance and asks, after the fact, whether $\hat{\tau}_\alpha^{(r)}$ happened to fall on or before the corresponding calendar date $M + kM$: it is read off the very same $\hat{\tau}_\alpha^{(r)}$, so it requires no extra simulation, but it answers a narrower, date-specific question rather than the open-ended “ever” question. Throughout this note, “sequential” and “pointwise” are used exactly in this sense: sequential = a property of the whole stopping time; pointwise = the stopping time’s indicator at one evaluation date $M + kM$. Both are computed for Scenario A (size, this section) and, in Section 7.2, for Scenario B (power) as well.*

Sequential statistic: empirical size. The classical Type-I error rate of the procedure *as deployed* is the probability that it ever alarms over the full monitoring horizon $T_{\text{total}} = 10M + 10$ of (22):

$$\widehat{\text{Size}}_\alpha(N, M, \text{factor}) = \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{\hat{\tau}_\alpha^{(r)} < \infty\}. \quad (23)$$

In the language of Remark 2, this is the *sequential* statistic for Scenario A: the indicator $\mathbf{1}\{\hat{\tau}_\alpha^{(r)} < \infty\}$ depends only on whether the stopping time is ever finite, with no reference to a fixed evaluation date. It is a single number per $(N, M, \text{factor}, \alpha)$ combination, with no k index, and it is the only quantity that should be compared directly to the nominal level α . It is the direct analogue of the “any alarm” rate $\hat{B}_\alpha$ used in Scenario B (30), which is likewise a sequential statistic.

Pointwise statistic: diagnostic CDF curve (not six independent sizes). For additional diagnostic detail, $\hat{\tau}_\alpha^{(r)}$ is also read off at six post-hoc window lengths $k \in \mathcal{K} = \{0.2, 0.3, 0.5, 1, 2, 3\}$:

$$A_\alpha^{(r)}(k) \equiv \mathbf{1}\{\hat{\tau}_\alpha^{(r)} \leq M + kM\}, \quad \hat{F}_\alpha(k) = \frac{1}{R} \sum_{r=1}^R A_\alpha^{(r)}(k). \quad (24)$$

In the language of Remark 2, $A_\alpha^{(r)}(k)$ is the *pointwise* indicator at evaluation date $M + kM$: unlike $\mathbf{1}\{\hat{\tau}_\alpha^{(r)} < \infty\}$ in (23), it does not ask whether the monitor ever alarms, only whether it has alarmed by the specific date $M + kM$ fixed in advance by k . Because all six indicators in (24) are read from the *same* $\hat{\tau}_\alpha^{(r)}$ within a given replication, they are nested: $k_1 < k_2 \Rightarrow A_\alpha^{(r)}(k_1) \leq A_\alpha^{(r)}(k_2)$. Averaging over r , $\hat{F}_\alpha(\cdot)$ is therefore the empirical CDF of the false-alarm time $\hat{\tau}_\alpha$ evaluated at six points, not six independent estimates of size. It should *not* be benchmarked against a flat α : for any finite k the true probability that the limiting Brownian bridge $W^0(\tau)$ of (17) crosses the boundary (20) within $\tau \in (1, 1 + k]$ is strictly less than α , approaching α only as $k \rightarrow \infty$ (since α is, by construction, the crossing probability over the infinite horizon $\tau \geq 1$ in (18)). The correct benchmark is instead the finite-horizon theoretical crossing probability

$$F_{\text{bridge}}(k; a_\alpha^2) = P\{\exists \tau \in (1, 1 + k] : |W^0(\tau)| \geq g(\tau; a_\alpha^2)\}, \quad (25)$$

which depends only on a_α^2 and k – not on (N, M, factor) – and is obtained from a separate, lightweight Monte Carlo simulation of the continuous-time process $W^0(\tau) = W(\tau) - \tau W(1)$ itself, with no panel data involved. Comparing $\hat{F}_\alpha(k)$ to $F_{\text{bridge}}(k; a_\alpha^2)$ isolates whether finite M , finite N , AR estimation error, and the pairwise aggregation of Section 4 cause the discrete panel statistic’s false-alarm timing to deviate from its own asymptotic limit – which is the operationally meaningful sense in which a finite-sample monitoring procedure can be said to be “well-sized.” A well-calibrated test should give $\widehat{\text{Size}}_\alpha \approx \alpha$ in (23) and $\hat{F}_\alpha(k) \approx F_{\text{bridge}}(k; a_\alpha^2)$ in (25); neither comparison should be made against the other benchmark.

7.2 Scenario B: power ($\delta \in \{1, 2, 3\}$)

A break of size δ is injected at

$$\tau(k) = M + \lfloor kM \rfloor, \quad k \in \mathcal{K}. \quad (26)$$

Here k is the break location, not a scoring window. Each (δ, k) combination requires its own set of replications because the data depend on (δ, k) . As in Scenario A, every replication produces exactly one stopping time $\hat{\tau}_\alpha^{(r)}$, and, exactly as in Remark 2, two distinct kinds of statistic are read off that same $\hat{\tau}_\alpha^{(r)}$ at no extra simulation cost: a *sequential* classification of the stopping time against the true break date $\tau(k)$, described first, and a *pointwise* indicator of whether the monitor has alarmed by a fixed evaluation date, described in Section 7.2.2. The two must not be conflated: the sequential statistics below are power and premature-alarm rates benchmarked against the actual break date $\tau(k)$, whereas the pointwise statistics evaluate the same stopping time at six calendar dates that have nothing to do with $\tau(k)$, just as in Scenario A.

7.2.1 Sequential statistics: premature alarm, detection, and delay

The stopping time $\hat{\tau}_\alpha^{(r)}$ is compared to the true break date $\tau(k)$ and classified as one of three outcomes:

$$P_\alpha^{(r)} \equiv \mathbf{1}\{\hat{\tau}_\alpha^{(r)} \leq \tau(k)\} \quad (\text{alarm before the break}), \quad (27)$$

$$D_\alpha^{(r)} \equiv \mathbf{1}\{\tau(k) < \hat{\tau}_\alpha^{(r)} < \infty\} \quad (\text{alarm after the break}), \quad (28)$$

$$B_\alpha^{(r)} \equiv P_\alpha^{(r)} + D_\alpha^{(r)} = \mathbf{1}\{\hat{\tau}_\alpha^{(r)} < \infty\} \quad (\text{any alarm}). \quad (29)$$

Each of $P_\alpha^{(r)}$, $D_\alpha^{(r)}$, and $B_\alpha^{(r)}$ is, in the sense of Remark 2, a *sequential* statistic: all three depend on $\hat{\tau}_\alpha^{(r)}$ only through where it falls relative to the single, replication-specific break date $\tau(k)$, with no fixed evaluation date involved. The Monte Carlo averages are

$$\hat{P}_\alpha = \frac{1}{R} \sum_r P_\alpha^{(r)}, \quad \hat{D}_\alpha = \frac{1}{R} \sum_r D_\alpha^{(r)}, \quad \hat{B}_\alpha = \hat{P}_\alpha + \hat{D}_\alpha. \quad (30)$$

$\hat{D}_\alpha$ is the power of the test — the probability of detecting the break after it occurs. $\hat{P}_\alpha$ is the premature alarm rate. $\hat{B}_\alpha$, the probability of any alarm over the full horizon regardless of timing, is the Scenario-B sequential statistic directly comparable to $\widehat{\text{Size}}_\alpha$ (23) from Scenario A.

For replications where $D_\alpha^{(r)} = 1$, the detection delay is

$$\text{Delay}_\alpha^{(r)} \equiv \hat{\tau}_\alpha^{(r)} - \tau(k) > 0. \quad (31)$$

Summary statistics (mean, standard deviation, median, 25th and 75th percentiles) are computed over the subset $\{r : D_\alpha^{(r)} = 1\}$. Premature alarms ($P_\alpha^{(r)} = 1$) are excluded from this calculation because they have a negative delay and measure a different phenomenon.

7.2.2 Pointwise statistic: alarm-by-date curve under a break

Exactly as in the diagnostic CDF of Scenario A (24), the same stopping time $\hat{\tau}_\alpha^{(r)}$ is also read off at the six fixed window lengths $k_0 \in \mathcal{K} = \{0.2, 0.3, 0.5, 1, 2, 3\}$, evaluated at calendar date $M + k_0M$:

$$B_\alpha^{(r)}(k_0) \equiv \mathbf{1}\{\hat{\tau}_\alpha^{(r)} \leq M + k_0M\}, \quad \hat{G}_\alpha(k_0) = \frac{1}{R} \sum_{r=1}^R B_\alpha^{(r)}(k_0). \quad (32)$$

The notation k_0 is deliberately distinct from the break location k in (26): k fixes *where the break occurs* and is part of the data-generating process, while k_0 fixes *when the monitor's alarm status is checked* and is applied only after $\hat{\tau}_\alpha^{(r)}$ has already been computed. In particular k_0 ranges over all six values in $\mathcal{K}$ for every fixed break location k , so a single $(N, \delta, k, M, \text{factor})$ combination produces six numbers $\hat{G}_\alpha(k_0)$, $k_0 \in \mathcal{K}$, one sequential triple $(\hat{P}_\alpha, \hat{D}_\alpha, \hat{B}_\alpha)$, and the associated delay statistics, all read off the same set of R replications at no extra simulation cost.

Remark 3 (Why the pointwise power curve is not the detection rate). $B_\alpha^{(r)}(k_0)$ in (32) does not distinguish whether $\hat{\tau}_\alpha^{(r)} \leq M + k_0M$ occurred before or after the true break date $\tau(k)$: it is the unconditional indicator that the monitor has alarmed by date $M + k_0M$, whatever the cause. It therefore mixes premature alarms and genuine detections together, exactly as $B_\alpha^{(r)}$ in (29) does over the full horizon, but evaluated at the fixed date $M + k_0M$ instead of over $(M, T_{\text{total}}]$. Consequently $\hat{G}_\alpha(k_0)$ is not an estimate of the detection rate $\hat{D}_\alpha$ at horizon k_0 , and it should not be read as one: when $k_0 < k$ (the evaluation date falls before the break), $\hat{G}_\alpha(k_0)$ can only reflect premature false alarms, since the break has not yet occurred, while when $k_0 \geq k$ it reflects a mixture of premature alarms that occurred before $\tau(k)$ and genuine detections that occurred between $\tau(k)$ and $M + k_0M$. As in Scenario A, the six values $\hat{G}_\alpha(k_0)$ are nested in k_0 because they are read from the same $\hat{\tau}_\alpha^{(r)}$, and they form an empirical CDF of the alarm time under the break, not six independent power estimates at six horizons.

7.3 Full design

Table 4 lists all parameter combinations.

The table reports simulation *combinations*, i.e. the number of distinct $(N, \delta, k, M, \text{factor})$ cells that must each be run for $R = 2500$ fresh replications; it is not a row count for the output tables. Within each combination, the sequential statistics (§7.2.1, or $\widehat{\text{Size}}_\alpha$ for Scenario A) occupy one output row, while the pointwise statistics (§7.2.2, or $\hat{F}_\alpha(k)$ for Scenario A) occupy six output rows, one per probe window $k_0 \in \mathcal{K}$, all computed from the same R replications at no extra simulation cost: 40 sequential rows and $40 \times 6 = 240$ pointwise rows for Scenario A; 720 sequential rows and $720 \times 6 = 4320$ pointwise rows for Scenario B — separately for each of the pairwise and pooled aggregation strategies of Section 4.

For $N = 30$, the random coefficients in Table 2 are redrawn on every replication, so the Monte Carlo averages integrate over both the coefficient distribution and the innovation sequences.

Table 4: Monte Carlo design.

Parameter	Scenario A (size)	Scenario B (power)
N	$\{10, 30\}$	$\{10, 30\}$
δ	0	$\{1, 2, 3\}$
k	— (no break is generated)	break location $\tau(k)$: $\{0.2, 0.3, 0.5, 1, 2, 3\}$
k_0 (pointwise probe)	evaluation date $M + k_0 M$: $\{0.2, 0.3, 0.5, 1, 2, 3\}$	evaluation date $M + k_0 M$: $\{0.2, 0.3, 0.5, 1, 2, 3\}$
M	$\{50, 80, 100, 200, 300\}$	$\{50, 80, 100, 200, 300\}$
Factor	$\{1, 2, 3, 4\}$	$\{1, 2, 3, 4\}$
R	2500	2500
Combinations	$2 \times 5 \times 4 = 40$	$2 \times 3 \times 6 \times 5 \times 4 = 720$
Total replications	10^5	1.8×10^6

8 Targeted Power Experiments

The design of Section 7 injects a level shift δ directly into every series. A companion design instead breaks the panel *indirectly*, through the common factor that loads onto the series, and asks whether the monitor still has power when the break works through the factor structure rather than through a direct level shift. Four experiments are run, crossing two break mechanisms (a break in the factor’s *volatility* versus a break in its *loading*) with two factor structures (a single factor shared by the whole panel versus factor-specific subgroups). All four reuse, unchanged, the AR pre-filtering of Section 3, the pairwise cross-sectional aggregation (14), and the CUSQ statistic and boundary of Section 5: only the data-generating process and the reported outcomes differ. The AR coefficients $\hat{\beta}_{AR}$, the pairwise coefficients $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$, and the means μ_i are again estimated once on $t = 1, \dots, M$ and held fixed thereafter. In all four experiments, k denotes only a break location, exactly as the break location k of Scenario B in Section 7.2 (it is *not* the pointwise probe window k_0 of Section 7.2.2, which is not computed for these four experiments): k is restricted to $k \in \{0.2, 0.3, 0.5\}$, and the break time is $\tau^*(k) = M + \lfloor kM \rfloor$ as in (26). Outcomes are reported with the sequential alarm/premature/detect/delay logic of (27)–(31), with $\tau(k)$ replaced by $\tau^*(k)$; to match the implementation’s column names, the Monte Carlo averages $\hat{B}_\alpha$, $\hat{P}_\alpha$, $\hat{D}_\alpha$ are denoted AR_α (alarm rate), PAR_α (premature alarm rate), and DR_α (detection rate), reported alongside the same five delay statistics. $R = 2500$ replications are run per combination, exactly as in Section 7.

8.1 Experiment 1: common-factor volatility break

Every series loads on factor 4 (the long-memory ARFIMA(0, 0.45, 0) factor of Table 3) with the same fixed loading as (1), $\gamma = 0.5$. No level shift is injected; instead, the i.i.d. $N(0, 1)$ innovations that drive factor 4 through fractional differencing, (36)–(37), switch standard

deviation from 1 to $\sqrt{3}$ at $\tau^*(k)$:

$$\varepsilon_{4,t} \sim N(0, 1) \text{ for } t < \tau^*(k), \quad \varepsilon_{4,t} \sim N(0, 3) \text{ for } t \geq \tau^*(k). \quad (33)$$

The idiosyncratic panel (Tables 1–2) is unaffected; only the variance of the common shock changes, so the break reaches every series solely through the factor loading.

8.2 Experiment 2: common-factor loading break

Each factor $j \in \{1, 2, 3, 4\}$ is now considered separately (no volatility break: the factor itself is generated exactly as in Table 3 throughout). Instead, every series' loading on $f_{j,t}$ jumps from $\gamma_{\text{pre}} = 0.1$ to $\gamma_{\text{post}} = 0.8$ at $\tau^*(k)$:

$$Y_{i,t}^* = Y_{i,t} + \gamma(t) f_{j,t}, \quad \gamma(t) = \begin{cases} 0.1, & t < \tau^*(k) \\ 0.8, & t \geq \tau^*(k), \end{cases} \quad i = 1, \dots, N. \quad (34)$$

Because each of the four factors is run as a separate combination, the parameter grid additionally ranges over factor $\in \{1, 2, 3, 4\}$, as in Scenario B of Section 7.

8.3 Experiments 3.1–3.2: heterogeneous (subgroup) factor structure

Experiments 8.1–8.2 load every series on the same single factor. Experiments 3.1 and 3.2 instead partition the N series into three disjoint subgroups G_2, G_3, G_4 , each loading on a *different* factor (f_2, f_3, f_4 respectively) and on no other factor (Table 5). Because no factor is shared by the whole panel, the panel-wide pooled signal is, by construction, only partially informative about a break confined to a single subgroup — this is the sense in which the factor structure is "weak" relative to Experiments 8.1–8.2.

Table 5: Subgroup composition for Experiments 3.1–3.2.

	Factor	$N = 10$	$N = 30$
G_2	f_2	$\{s_1, s_2, s_3\}$	$\{s_1, \dots, s_3\}$
G_3	f_3	$\{s_4, s_5, s_6\}$	$\{s_4, \dots, s_{12}\}$
G_4	f_4	$\{s_7, \dots, s_{10}\}$	$\{s_{13}, \dots, s_{30}\}$

For every replication, AR filtering is applied to the whole panel exactly as in Section 3 — each series keeps its own fixed $\text{AR}(p)$ coefficients regardless of subgroup membership. Two CUSQ tests are then computed from the *same* AR residuals: a full-panel test using (14) with all N columns, and, for each $g \in \{2, 3, 4\}$, a subgroup test using (14) restricted to the $|G_g|$ columns in G_g only. The closed form of Proposition 2 applies verbatim to any subset of columns, so the subgroup tests require no additional derivation; only the cross-sectional aggregation step changes, never the AR filtering. This lets each replication compare whole-panel monitoring against subgroup-specific monitoring for a break confined to one subgroup.

Experiment 3.1 (volatility break). Each subgroup’s own factor receives the same volatility break as (33) (std. $1 \rightarrow \sqrt{3}$ at $\tau^*(k)$), with a fixed loading $\gamma = 0.5$, applied only to the columns in that subgroup.

Experiment 3.2 (loading break). Each subgroup’s own factor has constant volatility, but the subgroup’s loading on it jumps from a subgroup-specific γ_{pre} to γ_{post} at $\tau^*(k)$ (Table 6), analogous to (34) but applied subgroup-by-subgroup with its own factor.

Table 6: Pre- and post-break loadings for Experiment 3.2.

Subgroup	Factor	γ_{pre}	γ_{post}
G_2	f_2	0.10	0.60
G_3	f_3	0.25	0.85
G_4	f_4	0.30	0.90

For both 3.1 and 3.2, the full-panel metrics are reported with prefix “full-” and each subgroup g ’s metrics with prefix “gg-”. In addition, for every pair of subgroups $(a, b) \in \{(2, 3), (3, 4), (2, 4)\}$ the gap in 5%-level detection rate is reported,

$$\Delta\text{DR}_{g_a, g_b} \equiv \text{DR}_{g_a, 5\%} - \text{DR}_{g_b, 5\%}, \quad (35)$$

summarizing whether a break carried by a more persistent (higher- d or higher- ϕ) factor is detected more or less reliably than one carried by a less persistent factor.

8.4 Design summary

Table 7 lists the four experiments’ combinations. M , factor, and the $N = 30$ random-coefficient redraw rule are exactly as in Table 4.

Table 7: Targeted power experiment design.

Experiment	N	M	k	Factor
1 (factor-4 vol. break)	$\{10, 30\}$	$\{50, 80, 100, 200, 300\}$	$\{0.2, 0.3, 0.5\}$	—
2 (loading break)	$\{10, 30\}$	$\{50, 80, 100, 200, 300\}$	$\{0.2, 0.3, 0.5\}$	$\{1, 2, 3, 4\}$
3.1 (subgroup vol. break)	$\{10, 30\}$	$\{50, 80, 100, 200, 300\}$	$\{0.2, 0.3, 0.5\}$	—
3.2 (subgroup loading break)	$\{10, 30\}$	$\{50, 80, 100, 200, 300\}$	$\{0.2, 0.3, 0.5\}$	—
Combinations: $30 + 120 + 30 + 30 = 210$; $R = 2500$ each; total replications $= 5.25 \times 10^5$.				

9 Computational Details

The full design requires 1.9×10^6 replications. Two changes to the implementation reduce computation time substantially without changing any statistic.

9.1 FFT-based fractional differencing

An ARFIMA process requires applying the fractional integration operator $(1 - L)^{-d}$, which has the moving-average representation

$$(1 - L)^{-d} = \sum_{j=0}^{\infty} w_j L^j, \quad w_j = \frac{\Gamma(j + d)}{\Gamma(j + 1) \Gamma(d)}. \quad (36)$$

Given a short-memory series Y_t , the fractionally integrated series is

$$X_t = \sum_{j=0}^t w_j Y_{t-j}, \quad t = 0, 1, \dots, T - 1. \quad (37)$$

Computing (37) by a direct loop costs $O(T^2)$ operations. The same result can be obtained by the discrete convolution

$$(w * Y) = \mathcal{F}^{-1}[\mathcal{F}(w) \odot \mathcal{F}(Y)], \quad (38)$$

where $\mathcal{F}$ is the DFT computed by the FFT algorithm, at cost $O(T \log T)$. The first T elements of $(w * Y)$ equal (37) exactly. For numerical stability, the weights are computed in log space:

$$w_j = \exp[\ln \Gamma(j + d) - \ln \Gamma(j + 1) - \ln \Gamma(d)], \quad (39)$$

which avoids overflow for large j . For $T = 4000$, this reduces the fractional differencing step from hundreds of milliseconds to under one millisecond per series.

9.2 Vectorized AR residuals

Equation (5) computes one residual per time period. A direct implementation loops over t and computes a dot product at each step, incurring $T - p$ separate Python function call overheads. The same result follows from a single matrix multiplication.

Lemma 1. *Let $t_0 = \max(8, p + 1)$ and $\mathcal{I} = \{t_0, \dots, T\}$. Define the lag matrix $Z \in \mathbb{R}^{|\mathcal{I}| \times p}$ by*

$$Z_{t,\ell} = Y_{i,t-\ell}, \quad \ell = 1, \dots, p, \quad t \in \mathcal{I}, \quad (40)$$

and let $y = (Y_{i,t})_{t \in \mathcal{I}}$. Then

$$(\hat{e}_{i,t})_{t \in \mathcal{I}} = y - Z \hat{\beta}_{AR}. \quad (41)$$

Proof. Row t of $Z \hat{\beta}_{AR}$ equals $\sum_{\ell=1}^p \hat{\beta}_{AR,\ell} Y_{i,t-\ell}$, which is the predicted value in (5). So $(y - Z \hat{\beta}_{AR})_t = \hat{e}_{i,t}$ for every $t \in \mathcal{I}$. $\square$

The matrix Z is formed once and the residuals for all periods are obtained in a single BLAS call. For $T \approx 3000$ and $p \leq 6$, this is about 30 times faster than the loop version, with identical numerical output.

9.3 Parallelisation

After both improvements, the time for one replication at $N = 30$, $M = 300$ falls from roughly 0.14 seconds to roughly 0.02 seconds, a reduction of about $8\times$. Across 1.9×10^6 replications, this brings the projected single-thread runtime from about 72 hours to about 9.5 hours.

The 760 parameter combinations in Table 4 are independent of one another, so each combination is assigned to one of 8 worker processes. Each worker is seeded from OS entropy to ensure independent random streams across processes. With 8-way parallelism, the expected wall-clock time is 1–1.5 hours.

The unpooled pairwise aggregation of Section 4 adds an $O(N^2M)$ Gram-matrix step per replication that was not present under pooling. For the grid used here ($N \leq 30$, $M \leq 300$) this step is small relative to data generation and AR filtering, so the per-replication timings above are not expected to change materially, but they should be re-benchmarked after the implementation update rather than assumed.

10 Summary

Scenario A is run once per combination (40 combinations $\times$ $R = 2500$ replications, exactly as before – this redesign reuses the existing stopping times and adds no simulation cost) and produces three outputs from the same $\hat{\tau}_\alpha^{(r)}$: a 40-row *sequential* table of $\widehat{\text{Size}}_\alpha$ (23), the single quantity to compare against the nominal $\alpha \in \{10\%, 5\%, 1\%\}$; a 240-row *pointwise* table (40 combinations $\times$ 6 probe windows k_0) of the diagnostic CDF $\widehat{F}_\alpha(k)$ (24); and a 6-row table of the theoretical benchmark $F_{\text{bridge}}(k; a_\alpha^2)$ (25), computed once from a standalone Brownian-bridge simulation and independent of (N, M, factor) . Scenario B likewise produces both kinds of output from the same $\hat{\tau}_\alpha^{(r)}$: a 720-row *sequential* table reporting $\widehat{B}_\alpha$, $\widehat{P}_\alpha$, $\widehat{D}_\alpha$, and the five delay statistics (27)–(31) for each α , plus a $720 \times 6 = 4320$ -row *pointwise* table of the alarm-by-date curve $\widehat{G}_\alpha(k_0)$ (32) for each α and each probe window $k_0 \in \mathcal{K}$ — with the caveat of Remark 3 that $\widehat{G}_\alpha(k_0)$ is not itself a detection rate. As in Section 4, both scenarios’ sequential and pointwise tables are produced separately for the pairwise and the pooled aggregation strategy.

In every replication, the CUSQ statistic uses $D_n = C_n/C_m - (n-2)/(m-2)$ from (15) and the boundary $g(n/m)$ from (20), both as specified in Wang and Hsiao (2013). Propositions 1 and 2 ensure that the (unpooled) cross-pair aggregate $\bar{e}_t$, with every pair (i, j) contributing its own fitted slope $\hat{\beta}_{ij}$, is nonetheless computed exactly, at cost $O(N^2M) + O(NT)$ instead of $O(N^2M) + O(N^2T)$ for a direct pair-by-pair evaluation. Lemma 1 and the FFT-based fractional-differencing step ensure that all residuals are numerically identical to a direct implementation.

References

- Brown, R. L., J. Durbin, and J. M. Evans (1975). Techniques for testing the constancy of regression relationships over time. *Journal of the Royal Statistical Society, Series B*, 37, 149–163.
- Chu, C.-S. J., M. Stinchcombe, and H. White (1996). Monitoring structural change. *Econometrica*, 64(5), 1045–1065.
- Inclan, C. and G. C. Tiao (1994). Use of cumulative sums of squares for retrospective detection of changes of variance. *Journal of the American Statistical Association*, 89(427), 913–923.
- Poskitt, D. S. (2007). Autoregressive approximation in nonstandard situations: the fractionally integrated and non-invertible cases. *Annals of the Institute of Statistical Mathematics*, 59, 697–725.
- Wang, C. S.-H. and C. Hsiao (2013). Real-time monitoring test for realized volatility. *Journal of Time Series Econometrics*, 5(1), 1–24.